Optimal Crop-Area Allocation for

Agricultural Resource Planning

A Linear Programming Approach using District-Level Crop Production Data (Government of India)

presented as part of **Operations Research – Review I**

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The Problem

Land is fixed; crops differ sharply in yield. Farmers allocate hectares by habit, not by comparing district-level yield records across crops.

- Yield per hectare varies widely across crops, districts and years.
- Land is limited – one crop's gain is another's loss.
- Historical records exist at district–season–year granularity, but are rarely used to re-plan next season's allocation.

Question: using actual historical yield records, how should land be re-allocated across crops to get the most out of a fixed area?

Objectives and Scope

Objectives

- Use the dataset's own `Yield` column (no derived ratios needed)
- Formulate crop allocation as an LP (max output; min land)
- Solve computationally (PuLP / SciPy `linprog`)
- Compare optimal vs. actual historical allocation

Scope

- Single state, single season, one representative year (or multi-year average)
- **No cost data exists in this dataset** – objective is output/land only, not profit or cost
- Only crops reported in the *same production unit* are pooled into one LP (see Why This Dataset)

Author, Year	Method	Contribution	Relevance
Wankhade & Lunge, 2012	LP	Resource allocation, 10 crops, Akola district	LP works on Indian district-level crop data
Majeke et al., 2020	LP	Land allocation, small-holder farm	LP plan beat farmer's original plan
Supriya et al., 2022	LP case study	Telangana crop plan, +3.75–15% returns	Realistic gains from LP crop planning
Paidipati & Padi, 2017	Two LP models	Land allocation: cost-min & revenue-max	Motivates a primal/dual pair of objectives

Gap

All cited studies use a one-time survey of a single district. None draw coefficients from a multi-year, multi-district *government* production record, as this project does.

Research Gap

Prior work: single-district survey data; assumes price is known, optimises profit directly.

Missing: an LP built entirely from an existing public dataset (reproducible), that stays honest about what the data lacks (no cost or price field here).

Our Positioning

Build the LP strictly from **area**, **production**, **yield** – output-maximisation and land-minimisation only. Cost/profit stays out rather than being approximated.

Dataset

Dataset	District-wise, season-wise Crop Production Statistics, India
Source	Ministry of Agriculture & Farmers Welfare, Govt. of India (data.gov.in), via Kaggle
Size	575,879 rows $\times$ 8 columns
Coverage	All India, district level, crop years 1997 – 2020-21

Fields per record

- `State_Name`, `District_Name` – location of cultivation
- `Crop_Year` – year of production (1997 – 2020-21)
- `Season` – Kharif, Rabi, Summer, Whole Year, Autumn, Winter
- `Crop` – crop name (Rice, Wheat, Sugarcane, Banana, Cotton, . . .)
- `Area (Hectares)`, `Production (Quintals)`, `Yield (Quintals/Hectare)` – Yield is already calculated as Production/Area

What the Data Actually Looks Like

State	District	Year	Season	Crop	Area (ha)	Prod.	Yield
A. & N. Islands	Nicobars	2000	Kharif	Arecanut	1,254.00	2,000.00	1.60
A. & N. Islands	Nicobars	2000	Kharif	Rice	102.00	321.00	3.15
A. & N. Islands	Nicobars	2001	Whole Year	Cashewnut	719.00	192.00	0.27
A. & N. Islands	Nicobars	2001	Whole Year	Coconut	18,190.00	64,430,000.00	3,542.06

Actual rows from the dataset (Yield column shown as given / directly computable). **Note the Coconut row:** its Production is counted in *nuts*, not tonnes, so its "Yield" is on a completely different scale from Rice or Arecanut. This is exactly why crops must be unit-checked before being pooled into one LP – see the next slide.

Learning from the last two reviews

Earlier feedback: don't propose a method the dataset can't support, and don't leave a data quirk undiscovered until it breaks the model. Both checks are applied here before formulating anything.

Data gives us

- Area → land constraint
- Yield (already computed) → output coefficient
- Multiple years/seasons → averaging, not guessing

Data does *not* give us – and a quirk it has

- No cost or price field → no profit/cost objective
- Some crops (e.g. Coconut) are counted in *nuts*, not weight → excluded when pooling crops into one output total

Government-sourced, district-level, multi-year – not the generic single-snapshot CSVs most submissions reuse.

What Will We Analyse?

Yield Analysis

- Yield/ha by crop, district, year
- Confirm which crops share a common production unit (weight) before pooling

State/District Comparison

- Area vs. production, district by district
- Best-yield crop per district, per season

Visuals planned: yield trend over years, top-crop bar charts per district, before/after allocation comparison once the LP is solved (Review II).

Optimization 1: Maximise Agricultural Output

Q1. Given a fixed land budget, how many hectares per crop maximise total production?

Decision	$x_{\text{Rice}}, x_{\text{Arecanut}}, \ldots$ = hectares given to each crop	**Constraints**	Land: $\sum_c x_c \leq L$
Objective	Maximise $\sum_c y_c x_c$		Bounds: $a_c \leq x_c \leq m_c$
Output	Optimal hectares & output per crop		Non-negativity: $x_c \geq 0$
OR Technique	Linear Programming		

y_c = the dataset's own `Yield` value for crop c (quintals/ha), averaged over available years for the chosen district/state. Only crops sharing the same production unit are included together.

Let the crops grown in the chosen state/district be $c \in \{\text{Rice, Arecanut}, \dots\}$ (restricted to crops reported in the same production unit), with y_c = average `Yield` for crop c from the dataset:

$$\text{Maximise } Z = y_{\text{Rice}}\, x_{\text{Rice}} + y_{\text{Arecanut}}\, x_{\text{Arecanut}} + \cdots = \sum_c y_c\, x_c$$

subject to

$$\sum_c x_c \le L \qquad \text{(total cultivable land)}$$

$$a_c \le x_c \le m_c \quad \forall c \qquad \text{(historical min/max area per crop)}$$

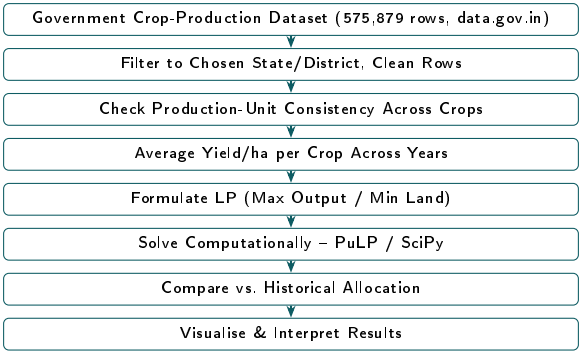
$$x_c \ge 0 \quad \forall c$$

Q2. To guarantee a minimum total production D, what's the smallest land footprint needed?

Decision	$x_{\text{Rice}}, x_{\text{Arecanut}}, \ldots$ = hectares given to each crop	**Constraints**	Target: $\sum_c y_c x_c \geq D$
Objective	Minimise $\sum_c x_c$		Bounds: $a_c \leq x_c \leq m_c$
Output	Least-land allocation vs. historical footprint		Non-negativity: $x_c \geq 0$
OR Technique	Linear Programming		

Same data, same y_c coefficients as Optimization 1 – objective and binding constraint swapped (the standard primal/dual pairing for a resource-allocation LP).

Project Workflow



Tools & Technologies

Data & Stats	Pandas + NumPy
Visualization	Matplotlib / Plotly
Optimization	PuLP, SciPy `linprog`
Interface	Streamlit (optional, Review III)

Assumptions & Limitations

1 – Production Units Must Match

Crops reported in count (e.g. Coconut, in nuts) rather than weight are excluded when pooling crops into a single output total – confirmed directly from the data, not assumed.

2 – Historical Averages

Yield/ha is averaged across the available years for the chosen state/district; year-to-year weather variation is not modelled in this base version.

3 – Output, Not Profit

No cost or price field exists in this dataset $\Rightarrow$ we optimise production/land, not profit.

4 – Not Modelled

Soil quality, irrigation, labour, crop rotation, price/demand shifts.

Work Plan and Timeline

Milestone	Deliverables
Review I	Problem, literature review, dataset check, LP formulation
Review II	State/district chosen, yields averaged, unit check done, both LP models solved computationally, vs. historical
Review III	Sensitivity analysis, multi-district comparison, final report

References

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Thank You

Questions & Discussion